COMPUTER NETWORKS LAB

B. Tech Semester **L T P C**
0 0 2 1

Subject Code: 22CS505PC

Co-requisite: A Course on "Computer Networks".

Prerequisites:
1. A course on "Programming for problem solving."
2. A course on "Data Structures"

Course Objectives
1. To understand the working principle of various communication protocols.
2. To analyze the traffic flow and the contents of protocol frames.
3. To interpret various routing algorithms.
4. To explore the concepts of data encryption and decryption.
5. To know congestion control and network and operating functionalities.

Course Outcomes:
After completion of this course, the students will be able to:
1. Implement data link layer farming methods.
2. Analyze error detection and error correction codes.
3. Design and analyze routing and congestion issues in network design.
4. Develop Encoding and Decoding techniques used in presentation layer
5. Work with different network tools

List of Experiments
1. Implement the data link layer framing methods such as character, character-stuffing and bit stuffing.
2. Write a program to compute CRC code for the polynomials CRC-12, CRC-16 and CRCCCIP
3. Develop a simple data link layer that performs flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
4. Implement Dijsktra's algorithm to compute the shortest path through a network
5. Take an example subnet of hosts and obtain a broadcast tree for the subnet.
6. Implement distance vector routing algorithm for obtaining routing tables at each node.
7. Implement data encryption and data decryption
8. Write a program for congestion control using Leaky bucket algorithm.
9. Write a program for frame sorting techniques used in buffers.
10. How to run N map scan.
11. Operating System Detection using N map.
12. Study of basic Network configuration commands and utilities to debug the network issues.

TEXT BOOKS:
1. Andrew S Tanenbaum, David. j. Wetherall, Computer Networks, 5th Edition, Pearson Education/PHI, 2011.
2. Behrouz A. Forouzan -Data Communications and Networking,4th Edition, Tata McGraw Hill, 2006.

REFERENCE BOOKS:

1. S. Keshav, An Engineering Approach to Computer Networks, Addison Wesley, 1st edition, 2010.
2. Larry L. Peterson and Bruce S. Davie: Computer Networks – A Systems Approach, 5th Edition, Elsevier, 2010.
3. Alberto Leon-Garcia and Indra Widjaja: Communication Networks -Fundamental Concepts and Key Architectures, 2nd Edition Tata McGraw-Hill, 2004.

WEB LINKS:

1. https://www.bits-pilani.ac.in/computer-networks-network-programming-lab/
2. What Is a Computer Network? Definition, Objectives, Components, Types, and Best Practices - Spiceworks
3. What is Computer Networking? - Networking in Computers Explained - AWS (amazon.com)
4. What is Computer Networking? - GeeksforGeeks
5. Prof Soumya Kanti Ghosh & Prof Sandip Chakraborty (youtube.com)
6. Introduction to Computer Networks (youtube.com)

CMR TECHNICAL CAMPUS
EXPLORE TO INVENT